

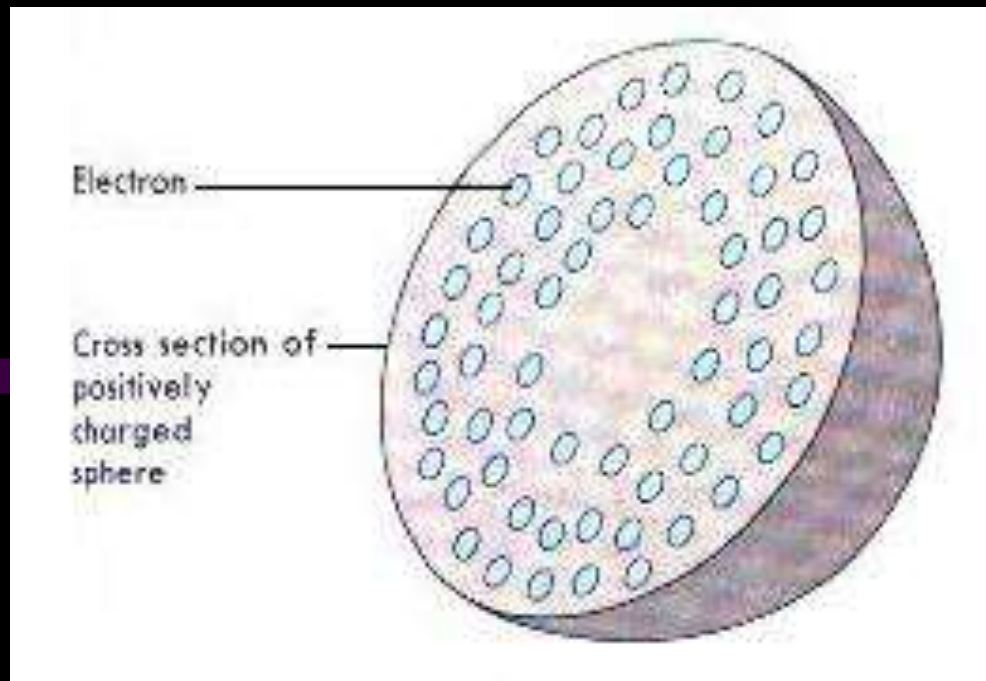
Bohr's atomic model



Democritus

Believed universe made of invisible units
called atoms

- Named them Atoms
- 400 BC
- Aristotle said “He’s a quack!!!”
- Took 2000 yrs to be proved right!



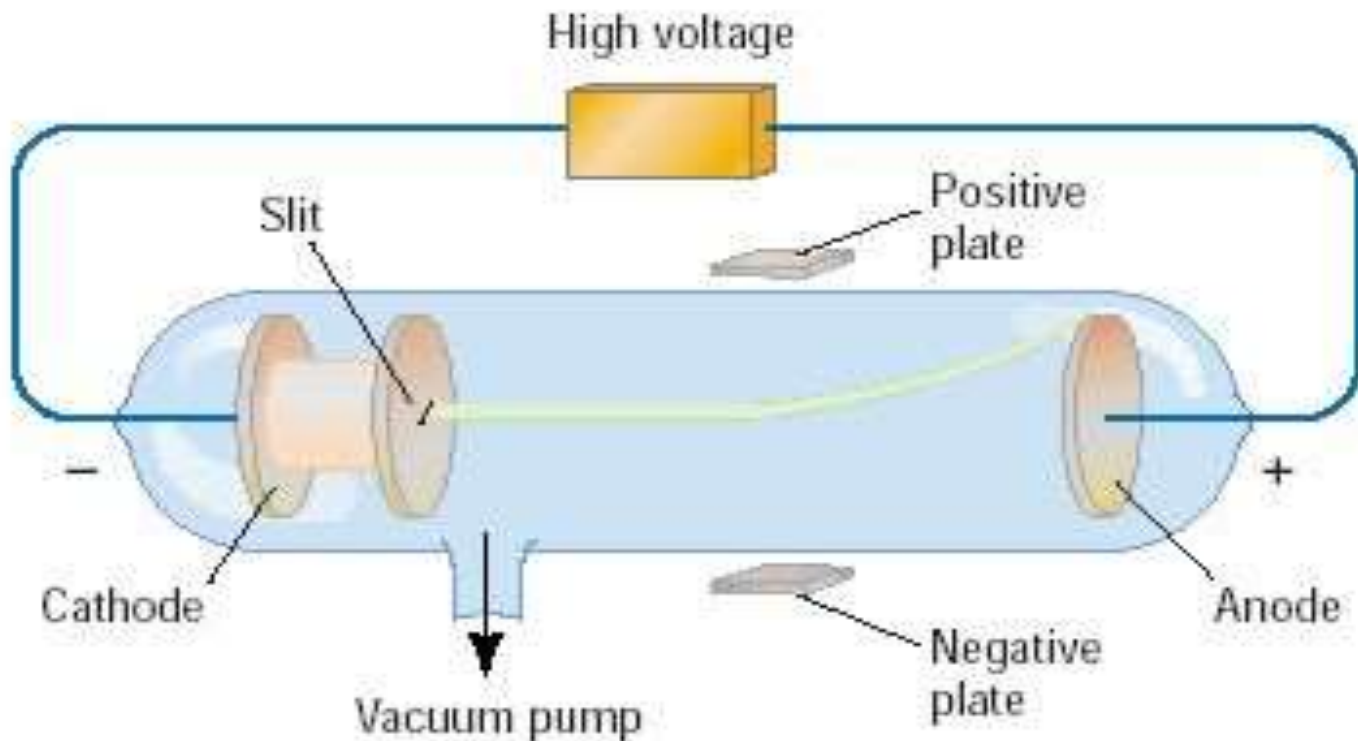
The Plum-Pudding

- **John Dalton (1803), England** - formed the atomic theory, which states that all matter is composed of tiny, indestructible particles called atoms that are all alike and have the same atomic weight. This is first scientific model. Atom is unbreakable part, it has some positive(+ve) and negative(-ve) charges.



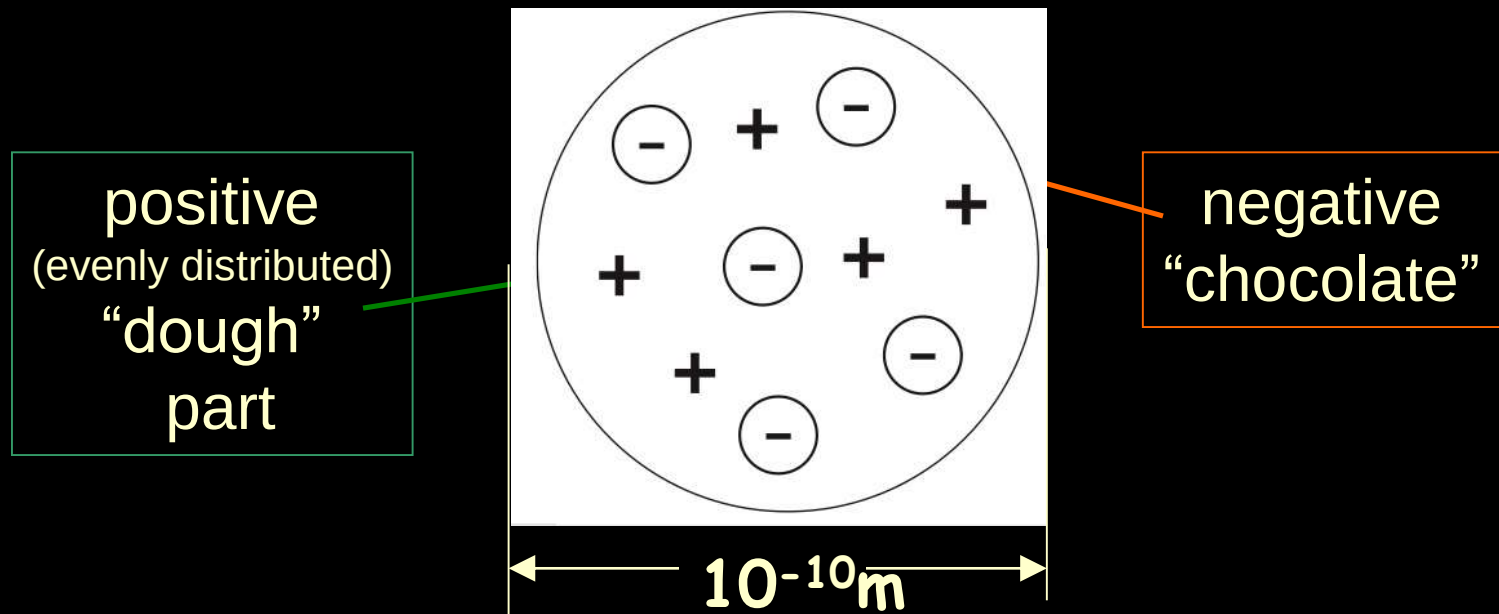
Discovery of the Electron

In 1897, J.J. Thomson used a cathode ray tube to deduce the presence of a negatively charged particle: the **electron**



Thomson: “Plum Pudding” or “Chocolate Chip Cookie” Model

- using available data on the atom, **J.J. Thomson** came up with the idea of having charges embedded with Dalton’s Billiard Balls
- Also used cathode ray experiment to discover the existence of the electron



note: this model kept Dalton’s key ideas intact

- **J. J. THOMSON:** He discovered the electron in 1891. The name of electron was suggested by Stony.
- Thomson says that atom is uniformly distributed. There is no empty space inside the atom. Electron are in static equilibrium with positive (+ve) charge.
- **DRAW BACKS:** No concept of nucleus. Most of alpha particles passed from gold foil, few are deflected through large angle, some are bouncing back.



Rutherford's Atomic Model

- Rutherford proposed that an atom has a very small positively charged core (nucleus) surrounded by the negative electrons.

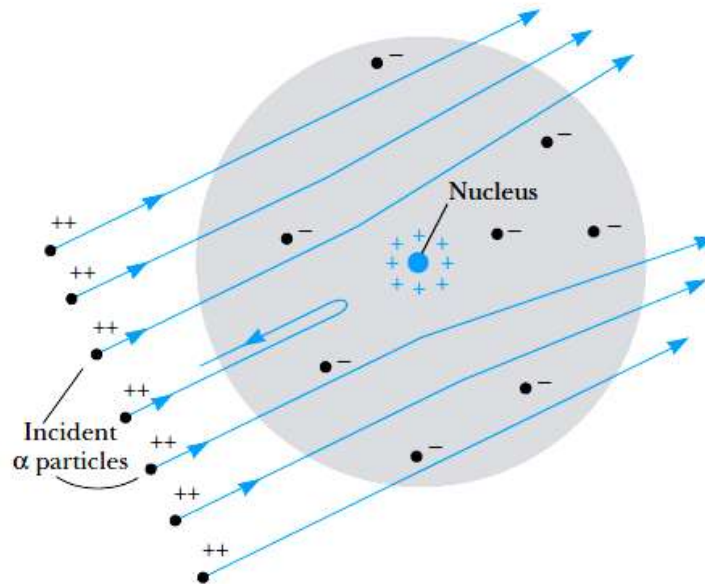
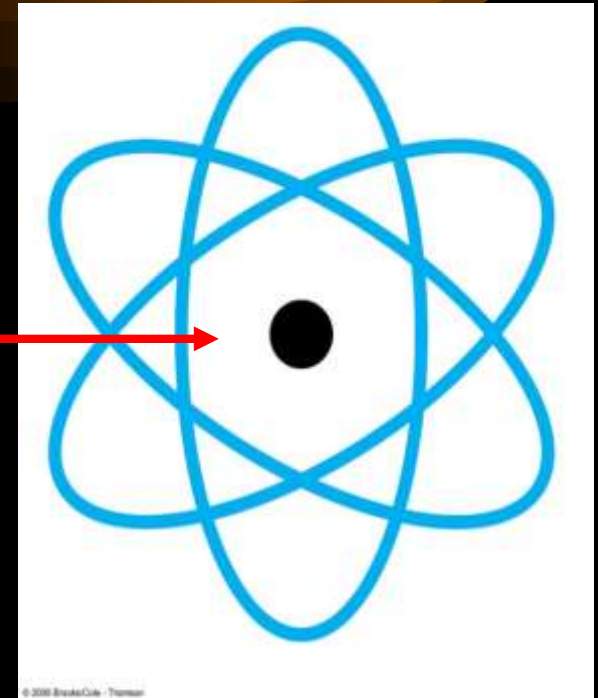


Figure 4.11 Scattering of α particles by a dense, positively charged nucleus.

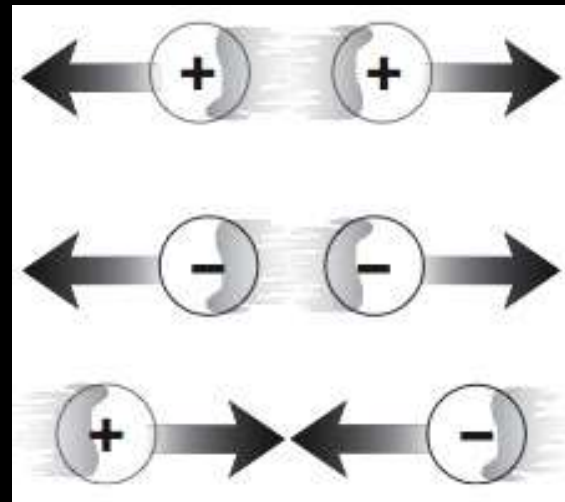


note: this model completely changed the definition of atom

- **RUTHER FORD (1909):** He used the results of his gold-foil experiment and discovered the nucleus, free space within an atom, that's why alpha particles passed easily from gold foil.
- Total charge on nucleus. Atom is like solar system. Nucleus consists protons and neutrons.
- **DRAW BACKS:** Total charge on nucleus, electrons moving around nucleus, during motion they radiating energy and will fall in nucleus.

New Evidence

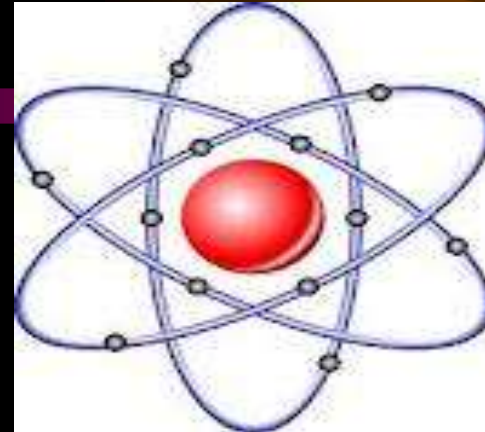
- during the 1900s evidence was discovered regarding charges:
 - atoms have positive (Rutherford's contribution) and negative (Thomson's contribution) parts
 - charges interact:



- as a result, revisions to Dalton's model had to be made

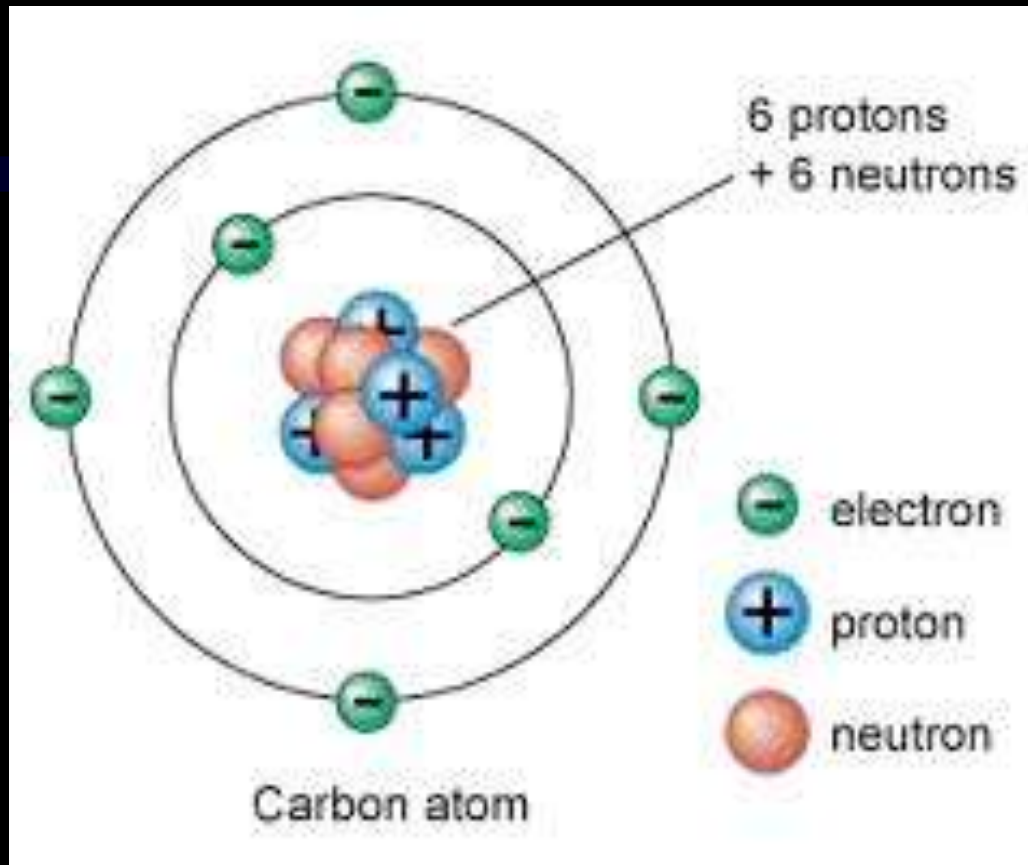
Niels Bohr

- Discovered that electrons exist in several distinct layers or levels
- “Jimmy Neutron Model”
- Travel around nucleus like planets travel around sun
- Electrons Orbit
- Electrons can jump between levels with energy being added/released



Bohr's Model of the Atom Bohr's model:

electrons orbit the nucleus like planets orbit the sun



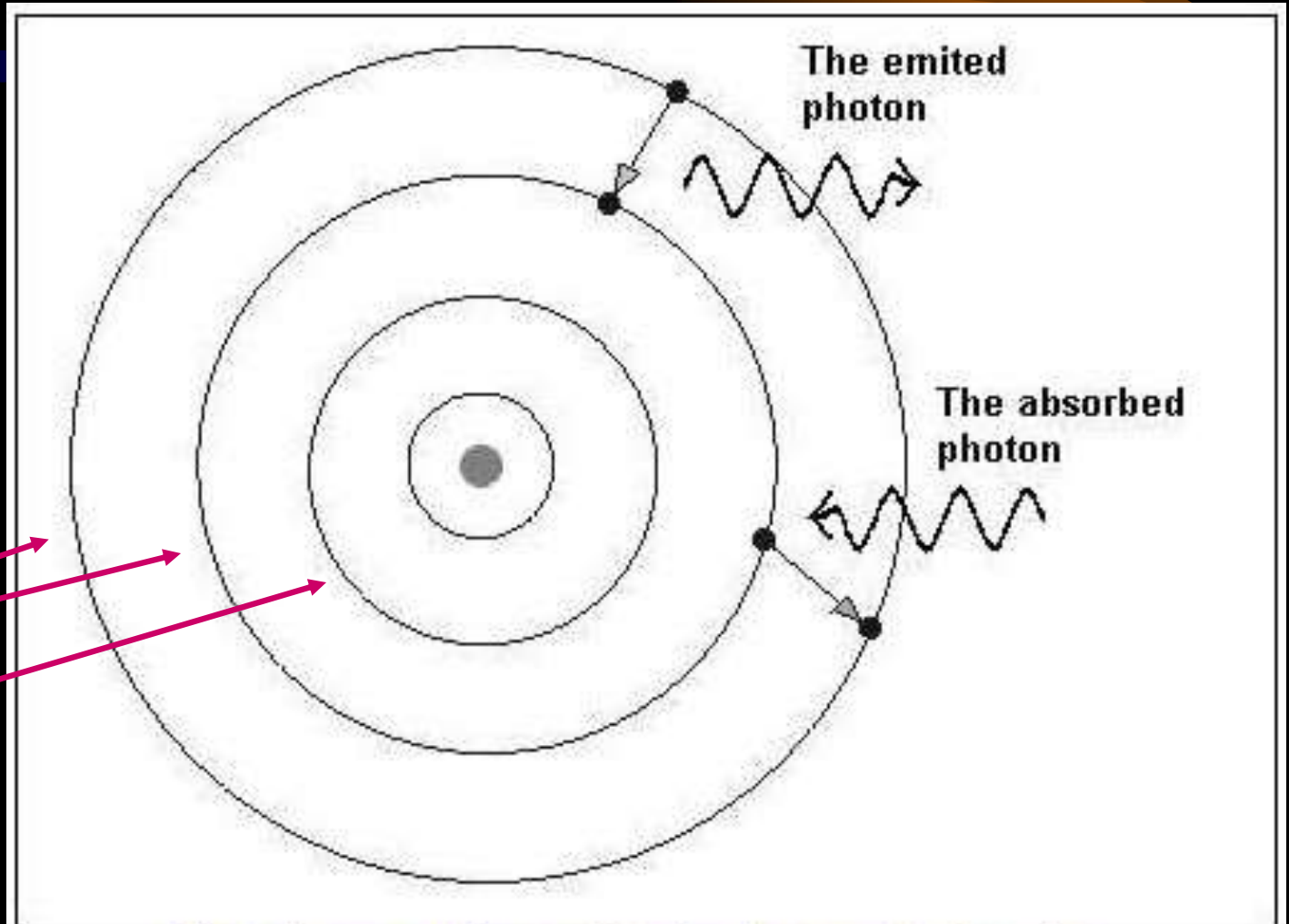
Bohr's Model of the Hydrogen Atom

- Bohr's model of the hydrogen atom is based on four assumptions. The first two are specific to his model. They are as follows:
 - The electron in a hydrogen atom moves in a circular orbit about the nucleus.
 - Only certain circular orbits are allowed. In these orbits the angular momentum $h/2\pi$ of the electron is quantized, just like energy is quantized in Einstein's photon model of light.

Bohr's Model of the Hydrogen Atom

- The next two assumptions are more general:
 - Electrons do not give off electromagnetic radiation when they are in an allowed orbit. Thus, the orbits are stable.
 - Electromagnetic radiation is given off or absorbed only when an electron changes from one allowed orbit to another. If the energy difference between the two allowed orbits is ΔE , then the frequency, f , of the photon that is emitted or absorbed is given by $\Delta E = hf$.

Bohr's idea



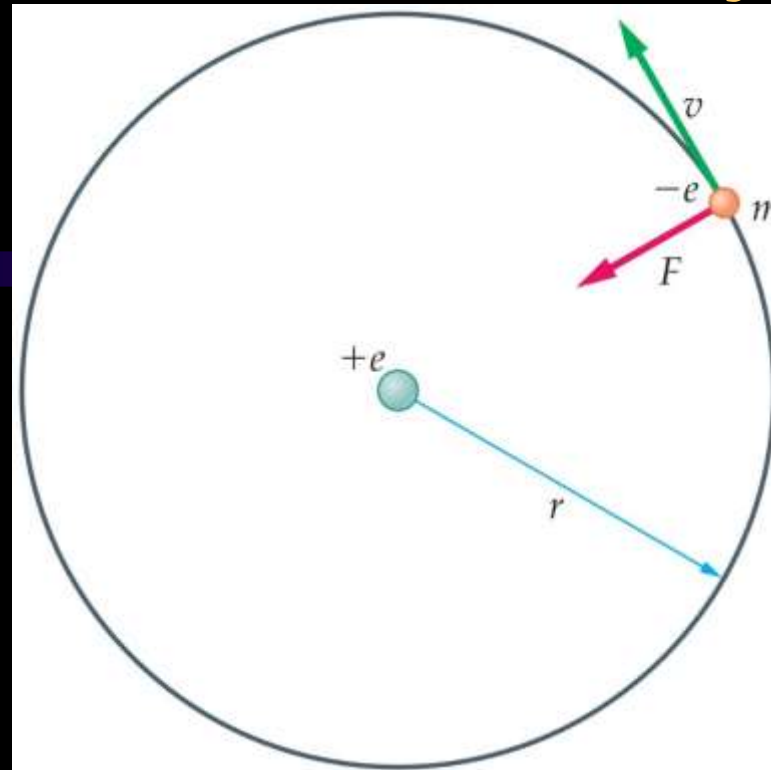
"Allowed"
orbits

The electron emits or absorbs the energy changing the orbits.

Bohr's Model of the Hydrogen Atom

- To summarize, Bohr's model retains the classical picture of an electron orbiting a nucleus. It also adds the quantum requirements that only certain orbits are allowed and no radiation is given off from these orbits.
- To determine the allowed Bohr orbits—those with specific radii and specific energies—we apply two conditions.
- First, the electron moves in a circular orbit of radius r and speed v ,

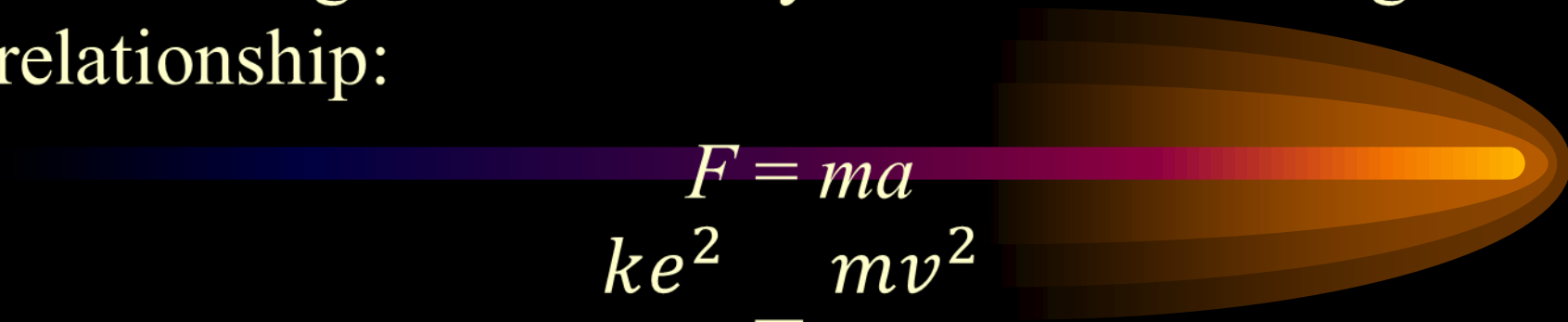
Radius of Bohr's Model of the Hydrogen Atom



- As a result, the electron experiences a centripetal acceleration toward the nucleus of magnitude $a = v^2/r$.
- From Newton's second law of motion, we know that a force, $F = ma$, is required to produce an acceleration. In this case, the force is the electrostatic force $F = ke^2/r^2$.

Radius of Bohr's Model of the Hydrogen Atom

- Combining these results yields the following relationship:


$$F = ma$$
$$\frac{ke^2}{r^2} = \frac{mv^2}{r}$$

$$v^2 = \frac{ke^2}{mr} \longrightarrow 1$$

Since the electron moves with a speed v in a circular path of radius r , its angular momentum is $L = mvr$.

According to Bohr's postulate

$$L = nh/2\pi$$

- $mvr = \frac{nh}{2\pi}$

- $v = \frac{nh}{2\pi mr}$

- Squaring both sides

- $v^2 = \frac{n^2 h^2}{4\pi^2 m^2 r^2}$

- Put the value of v^2 in equation 1

- $\frac{n^2 h^2}{4\pi^2 mr} = ke^2$

- $\frac{n^2 h^2}{4\pi^2 mke^2} = r$

- $r_n = \frac{h^2}{4\pi^2 mke^2} \times n^2 \longrightarrow 2$

Bohr's Model of the Hydrogen Atom

- Combining the force and angular momentum equations allows us to solve for the radii of allowed orbits. The result is

$$r_n = (h^2/4\pi^2mke^2)n^2 \quad n = 1, 2, 3, \dots$$

- Substituting the known values for h , π , m , k , and e yields the following result:

- $r_n = 0.529 \times 10^{-10} \times n^2$

Orbital Radius of an Electron in a Hydrogen Atom

orbital radius
of electron $= (5.29 \times 10^{-11} \text{ m}) \times (\text{quantum number})^2$

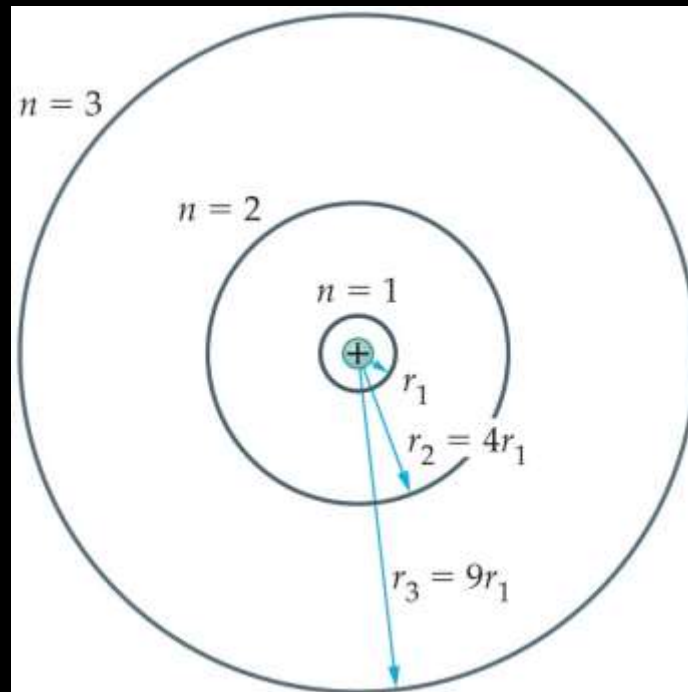
$$r_n = (5.29 \times 10^{-11} \text{ m})n^2 \quad n = 1, 2, 3, \dots$$

Bohr's Model of the Hydrogen Atom

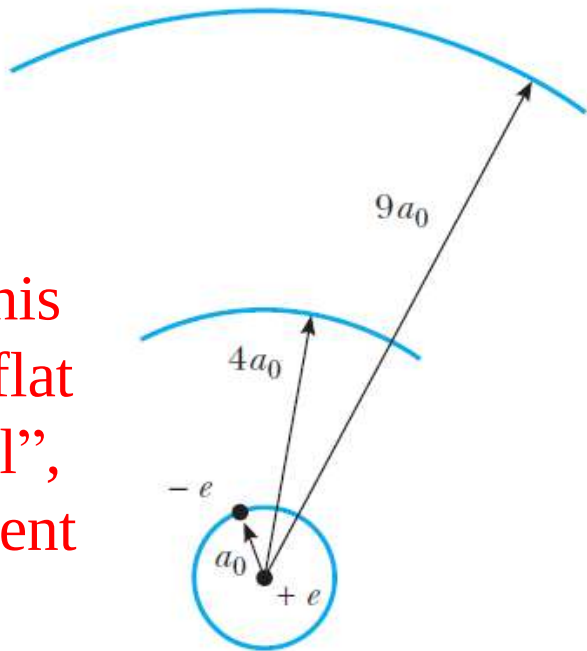
- The radius of the first Bohr orbit, which corresponds to $n = 1$, is

$$r_1 = 5.29 \times 10^{-11} \text{ m}$$

- This is known as the Bohr radius. It sets the typical size of a hydrogen atom. The higher orbits in the Bohr model increase in radius as n^2 , as indicated in the figure below.



Radii of Bohr orbits in hydrogen



$$r_n = n^2 a_0 \quad n = 1, 2, 3, \dots$$

Figure 4.23 The first three Bohr orbits for hydrogen.

$$r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{me^2} \equiv n^2 a_0$$

radius of the hydrogen atom for stationary states, $n = 1, 2, 3, \dots$

The smallest diameter of the hydrogen atom is two times the Bohr radius $\approx 1 \text{ \AA} = 0.1 \text{ nm}$

a_0

- $n = 1$ gives its lowest energy state (called the “ground state”), there is formally no E_0 since quantum numbers start with 1 and are positive integers

$$E_n = -\frac{ke^2}{2a_0} \left(\frac{1}{n^2} \right) \quad n = 1, 2, 3, \dots$$

But some books may call it E_0 nevertheless, E_1 is also called 1 Rydberg $\approx 13.6 \text{ eV}$, needs to be divided by h and c in order to get

Note that this is a “flat model”, apparently 2D

Bohr's Model of the Hydrogen Atom

- The total energy of the hydrogen atom is also quantized. In fact, a straightforward calculation, combining both the kinetic energy ($\frac{1}{2}mv^2$) and potential energy ($-ke^2/r$), shows that the total energy of the n th Bohr orbit is
- From equation 1

- $mv^2 = \frac{ke^2}{r}$, so

- $K.E = \frac{ke^2}{2r}$

- $P.E = W = \frac{ke^2}{r^2} r \cos(180^\circ)$

- $P.E=W = -\frac{ke^2}{r} \quad \therefore \cos(180^\circ) = -1$

Total energy is sum of K.E plus P.E

$$E = K.E + P.E$$

$$E = \left(\frac{ke^2}{2r}\right) + \left(-\frac{ke^2}{r}\right)$$

$$E = \frac{ke^2}{2r} - \frac{ke^2}{r}$$

$$E = \frac{ke^2 - 2ke^2}{2r}$$

$$E = -\frac{ke^2}{2r} \longrightarrow 3$$

Put the value of r from equation 2

$$E = - \frac{ke^2}{2 \left(\frac{n^2 h^2}{4\pi^2 m k e^2} \right)}$$

$$E = - \frac{ke^2 4\pi^2 m k e^2}{2 n^2 h^2}$$

$$E = - \frac{k^2 e^4 4\pi^2 m}{2 h^2} X \frac{1}{n^2}$$

OR

$$E_n = - \frac{e^4 m}{32\pi^2 \epsilon_o^2 \hbar^2} X \frac{1}{n^2}$$

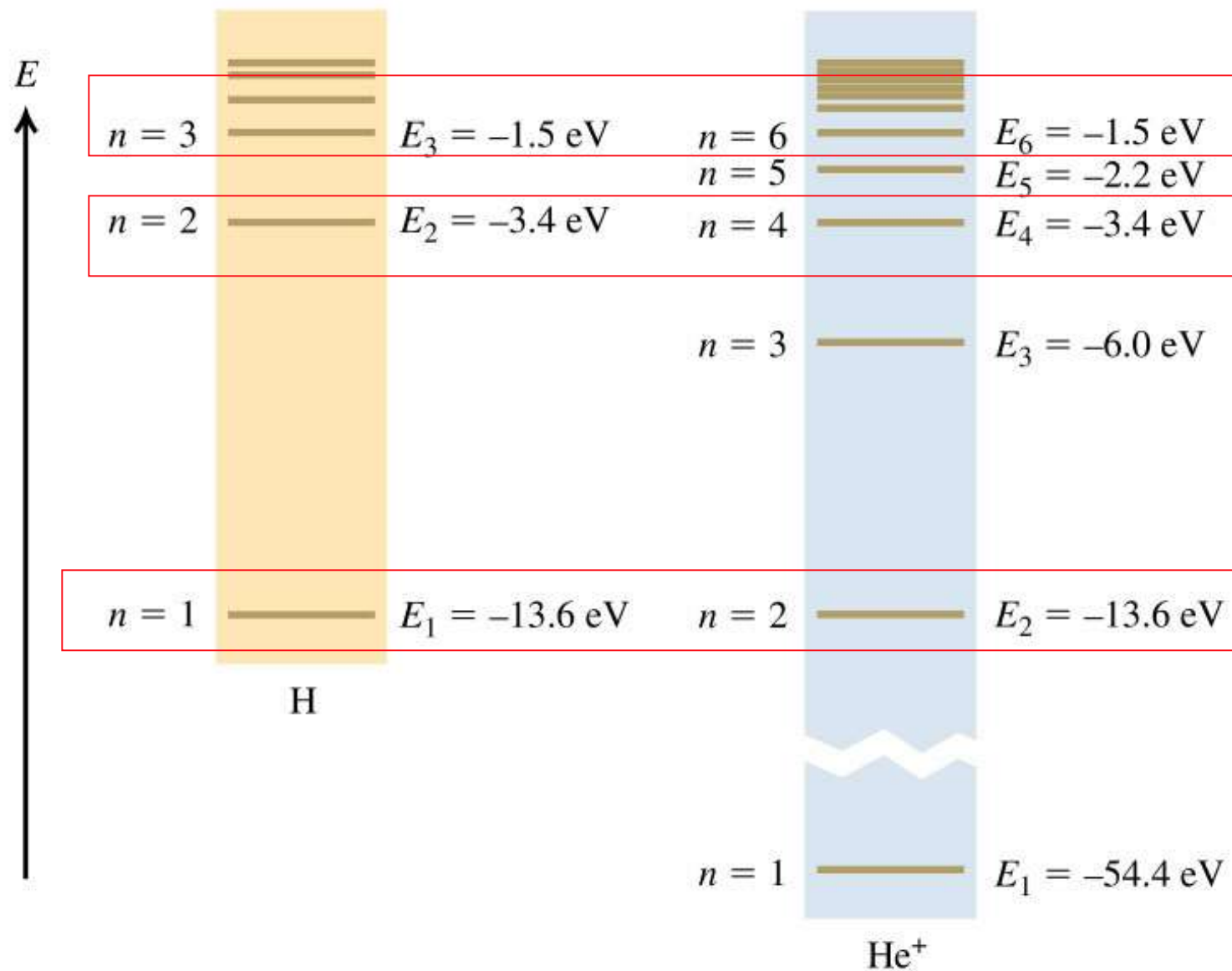
Substituting the known values for h , π , m , k , and e yields the following result:

Energy of a Hydrogen Atom

$$\text{total energy of a hydrogen atom} = - (13.6 \text{ eV}) \times \frac{1}{(\text{quantum number})^2}$$

$$E_n = - (13.6 \text{ eV}) \frac{1}{n^2} \quad n = 1, 2, 3, \dots$$

Ionized Helium is also a 1-electron atom



Bohr's Model of the Hydrogen Atom

GUIDED Example 25.3 | The Hydrogen Lineup (Continued)

Hydrogen Spectrum

Solution

- 1 Calculate the energy of the $n = 5$ state of hydrogen. This is the initial energy:

$$E_5 = -(13.6 \text{ eV}) \frac{1}{5^2} = -0.544 \text{ eV}$$

- 2 Calculate the energy of the $n = 2$ state of hydrogen. This is the final energy:

$$E_2 = -(13.6 \text{ eV}) \frac{1}{2^2} = -3.40 \text{ eV}$$

- 3 Use the energies from Steps 1 and 2 to calculate the magnitude of the change in energy. Convert the result to joules:

$$\begin{aligned} |\Delta E| &= |E_2 - E_5| \\ &= |-3.40 \text{ eV} - (-0.544 \text{ eV})| \\ &= 2.86 \text{ eV} \\ &= (2.86 \text{ eV}) \left(\frac{1.60 \times 10^{-19} \text{ J}}{1 \text{ eV}} \right) \\ &= 4.58 \times 10^{-19} \text{ J} \end{aligned}$$

Math HELP
Unit
Conversion
See
Lesson 1.3

- 4 Use the equation $\lambda = hc/|\Delta E|$ to find the wavelength for this spectral line:

$$\begin{aligned} \lambda &= \frac{hc}{|\Delta E|} \\ &= \frac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(3.00 \times 10^8 \text{ m/s})}{4.58 \times 10^{-19} \text{ J}} \\ &= 4.34 \times 10^{-7} \text{ m, or } 434 \text{ nm} \end{aligned}$$

Insight

The same procedure can be used to find the wavelength of any spectral line in hydrogen.

Frequency of the Hydrogen Atom

- According to Bohr's postulate

- $E = h\nu = E_i - E_f$

- $$h\nu = \left(-\frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} X \frac{1}{n_i^2} \right) - \left(-\frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} X \frac{1}{n_f^2} \right)$$

- $$h\nu = -\frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} X \frac{1}{n_i^2} + \frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} X \frac{1}{n_f^2}$$

- $$h\nu = \frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} X \frac{1}{n_f^2} - \frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} X \frac{1}{n_i^2}$$

- $$h\nu = \frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

- $\vartheta 2\pi\hbar = \frac{e^4 m}{32\pi^2 \epsilon_0^2 \hbar^2} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \therefore \hbar = \frac{h}{2\pi} = 2\pi\hbar = h$
 - $\vartheta = \frac{e^4 m}{64\pi^3 \epsilon_0^2 \hbar^3} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$
 - $\vartheta = \frac{c}{\lambda} = \frac{e^4 m}{64\pi^3 \epsilon_0^2 \hbar^3} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$
 - $\vartheta = \frac{1}{\lambda} = \frac{e^4 m}{64\pi^3 \epsilon_0^2 \hbar^3 c} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$
 - $\vartheta = \frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$
 - R_H is Rydberg constant
 - $R_H = 1.09 \times 10^7 \text{ m}^{-1}$
- 